

Programming Assignment#1: Sensor Monitoring System

Objective: Practice using loops, switch statements, and enumerations.

1 Overview

Write a program that simulates a simple sensor monitoring system. The program should repeatedly display the following menu:

```
1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit
```

Requirements

- Use an `enum` to represent the sensor types.
- Use a `while` loop to repeatedly display the menu.
- Use a `switch` statement to process menu selections.
- Store the most recent reading for each sensor.

2 Code Documentation

The very first lines in *all* of your lab project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, should be the name of the assignment and your name, e.g:

```
printf("Programming Assignment #1 Sensor Monitoring System - Joe Student\n");
```

The next line of output should be `printf("EGRE 245: Introduction to Programming Using C, Due: March 16, 2026\n");`

3 Sample Run

Your output should match the sample run below.

```
Programming Assignment #1: Sensor Monitoring System -- Mark Schwitzerlett
EGRE 245: Introduction to Programming Using C, Due: March 16, 2026

1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit

Enter Your Selection: 1
Enter temperature: 25.4

1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit

Enter Your Selection: 2
Enter voltage: 3.3

1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit

Enter Your Selection: 3
Enter current: 0.8

1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit
```

```
Enter Your Selection: 4
Temperature: 25.40 C
Voltage: 3.30 V
Current: 0.80 A

1 Enter temperature
2 Enter voltage
3 Enter current
4 Display readings
5 Exit

Enter Your Selection: 5

Program terminated.

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Please name your file `Assignment1_FirstName_LastName.c`.

Due date: This assignment is due Wednesday, March 18 at 11:59pm.

5 Academic Integrity

Any work submitted should be the student’s original work. Submitting code written by AI using topics not covered in class is strictly prohibited. Read the VCU Academic Misconduct standard [here](#).